PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: MyGov-Guard AI

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Project Name: MyGov-Guard AI

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysians are increasingly targeted by phishing scams that impersonate government agencies through emails and physical letters. At the same time, even legitimate government messages are often difficult to understand due to complex and formal language. This creates two main problems which are people may fall for scams and people may not understand real messages and take the wrong action.

This is especially challenging for elderly users and those who are not familiar with digital services.

- **Target Domain:** Domain 1: AI Systems & Agentic Workflow Automation
- **Proposed Solution Summary:** MyGov-Guard AI acts as a secure bridge between citizens and government services. Users can input text directly such as copy and pasting an email, upload a screenshot, or take a photo of a physical letter. The system will:

    - Check whether the message is real or a scam
    - Translate complex content into simple language
    - Guide users with clear, step-by-step actions using verified official links

The goal is to help users understand and respond correctly without confusion or risk.

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2. Background & Business Objective

- **Background of the Problem:** As government services such as LHDN, JPJ, and PADU become more digital, scams are also becoming more advanced. Currently:
  - Users must manually check if a message is real
  - Understanding government messages requires experience or knowledge
  - There is no simple tool that combines verification + explanation + action

  This makes the process slow, confusing, and risky.

- **Importance of Solving This Issue:** Solving this problem can:
  - Reduce financial loss caused by scams
  - Help users make correct decisions faster
  - Improve access to digital government services for all groups

It is especially important for vulnerable users who may not know how to verify information.

- **Strategic Fit / Impact:** This project supports Malaysia's digital transformation goals and anti-scam efforts. It helps turn a confusing and manual process into a simple, guided, and secure experience.

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3. Product Purpose

3.1. Main Goal of the System

To provide a safe and easy-to-use assistant that helps Malaysians verify, understand, and act on government-related messages.

3.2. Intended Users (Target Audience)

The MyGov-Guard AI platform is designed as a "digital shield" for Malaysian citizens. It follows a tiered approach to support users with different levels of digital skills and exposure to scam risks.

Primary Focus Segment (Most Vulnerable Group):

The main target users are the elderly and B40 communities. These groups often face challenges in using digital technology and may have difficulty understanding complex government-related language such as legal, tax, or banking terms. They are also more likely to be targeted by scammers using fear tactics, such as fake messages from agencies like LHDN or PDRM. As a result, they face a higher risk of financial loss. For this group, the application acts as a safety support tool by translating complicated or threatening messages into simple explanations in their preferred language.

Secondary Segment (General Users):

The platform is also designed for the general public, including working adults, students, and digitally active users. Nowadays, scams are becoming more advanced, using realistic-looking documents and messages. Even users with good digital knowledge may find it difficult to identify these scams. For this group, the application provides a quick and convenient way to check suspicious content, helping them save time and reduce uncertainty.

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4. System Functionalities

4.1. Description

The system works as a multi-step AI workflow. It takes unstructured input such as images or text, processes it, and produces a clear and structured output that users can easily follow.

4.2. Key Functionalities

- **Scam Detection Layer:** Checks sender details and message content to identify possible scams using known patterns.
- **Smart Translation Engine:** Converts complex government or legal language into simple, easy-to-understand explanations.
- **Action Guide (SOP Navigator):** Provides step-by-step instructions on what the user should do next, including official and safe links.
- **Multi-Modal Input Support:** Supports screenshots, photos (OCR), and text input.
- **Audio Accessibility Engine:** Synthesizes localized, high-quality speech to read the scam analysis summaries aloud, ensuring the application remains fully accessible to elderly citizens and the visually impaired.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

We use Gemini 2.5 Flash Processing because it:

- Supports both text and image understanding
- Handles multiple languages (English, Malay, Chinese)
- Works well with real-world inputs like screenshots and scanned physical letters

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4.3.2. Prompting Strategy

We use a multi-step approach with different roles:

- Agent 1: Scam detection (security focus)
- Agent 2: Translation (simplifying content)
- Agent 3: Action guide (step-by-step instructions)

Why this fits our use case:

Separating these steps ensures that verification is done first before any translation or action is generated. This reduces the risk of processing malicious content and improves overall reliability.

4.3.3. Context & Input Handling

- **Unstructured Input Handling**
  The system processes unstructured inputs such as images and PDF documents using Optical Character Recognition (OCR). Tesseract OCR is used for image-based text extraction, while Poppler is used to process PDF files by converting multi-page documents into sequential images. The extracted text is then sent to the Google Vertex AI (Gemini 2.5 Flash) model, which analyzes the content and filters out irrelevant information based on predefined prompt instructions.

- **Input Capacity**
  The application allows users to upload either a single image or a full PDF document in one request. By leveraging the large context window of Gemini 2.5 Flash, the system is able to process long documents such as multi-page government letters or legal notices without requiring strict character limits.

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- **Handling Edge Cases & Oversized Inputs**

  The system is designed to handle complex and large inputs efficiently. Instead of splitting the document into smaller parts, the entire extracted text is processed at once. This ensures that important details, such as hidden sender information or fine print, are not lost during analysis.

  In situations where the input is unreadable, too large, or when there are API connection issues, the system activates a safety mechanism. Rather than failing or crashing, the system responds with a clear fallback message. For example, users may be advised to contact official services such as the NSRC 997 hotline for manual verification. This approach ensures that user safety and system reliability are maintained at all times.

5. User Stories & Use Cases

- **User Stories:**
  - As a senior citizen, I want to check if a message is a scam before taking action.
  - As an SME owner, I want to quickly understand an LHDN or JPJ letter and know what to do next.
- **Use Cases (Main Interactions):**

  1. Verification Flow: User uploads image → system checks → result shown (Scam/Safe)

  2. System extracts key details → explains content → provides clear next steps

6. Features Included (Scope Definition)

- Scam detection for emails and physical letters
- Simplified translation of government messages
- Image-to-text processing (OCR)
- Text-to-speech processing
- Step-by-step action guidance

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7. Features Not Included (Scope Control)

- No direct integration with internal government databases
- No legal or financial advisory services
- No in-app payment system

8. Assumptions & Constraints

- **Cost Constraint:** Image processing can be expensive. The solutions are cache common government letter formats (e.g. Borang B standard layouts) and reduce repeated processing.
- **Technical Constraint:** The system depends on updated scam patterns and government procedures.
- **Performance Constraint:** Processing images may take a few seconds. The solution is to provide loading indicators and feedback.
- **User Input Constraint:** If images are unclear, users will be asked to retake them.

9. Risks & Questions Throughout Development

- **1. False Detection**
  - Risk: Real messages marked as scams
  - Solution: Use confidence levels and always provide official verification options
- **2. Data Privacy**
  - Ensure that user data (e.g. IC numbers) is not stored or used for training
- **3. Language Complexity**
  - Handling mixed language such as Malay and English may be challenging

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